

Wei Lin Chen

Research Scientist, Machine Learning

(650) 555-0184 | Palo Alto, CA | weilin.chen@example.com | linkedin.com/in/weilinchen | github.com/wlchen

SUMMARY

Machine learning research scientist with a PhD and 6 years turning research into shipped product. First-author work at NeurIPS and ICML on efficient inference; the resulting distillation method cut serving cost 47% for a model used by 30M users. Comfortable owning a problem end to end, from literature review to production rollout.

SKILLS

Research: Model compression, Distillation, Efficient inference, Representation learning, Evaluation design

Languages: Python, C++, CUDA, SQL

Frameworks: PyTorch, JAX, TensorFlow, Hugging Face Transformers, Triton

Infrastructure: Ray, Kubernetes, AWS (SageMaker, EC2), Slurm, Weights & Biases

Methods: Distributed training, Mixed precision, Quantization, A/B testing, Statistical analysis

EXPERIENCE

Research Scientist

Sep 2021 – Present

Applied AI Lab, Large Technology Company

Palo Alto, CA

- Developed a layer-wise distillation method that reduced inference cost 47% at equal accuracy on a ranking model serving 30M monthly users.
- Published 4 peer-reviewed papers (NeurIPS, ICML, EMNLP) and shipped 2 of the methods into production systems.
- Led a 5-person cross-functional effort to build the internal evaluation harness now used by 12 modeling teams.
- Cut distributed training time 3.2x by migrating to JAX with sharded data parallelism across 256 TPU cores.

Applied Scientist

Jul 2019 – Aug 2021

E-Commerce Search Team

Seattle, WA

- Built a query-understanding model that improved search relevance 9% NDCG and drove a 2.3% lift in conversion.
- Designed the offline-to-online evaluation pipeline that reduced failed A/B launches from 40% to 12%.
- Mentored 3 PhD interns; 2 papers accepted and both converted to full-time offers.

PROJECTS

slimkit | Python, PyTorch

github.com/wlchen/slimkit

- Open-source model compression toolkit implementing 6 distillation and quantization methods; 2.1K GitHub stars.

EDUCATION

Ph.D. in Computer Science

Stanford University

Jun 2019

Stanford, CA

B.S. in Mathematics and Computer Science

University of Illinois Urbana-Champaign

May 2014

Urbana, IL

PUBLICATIONS

- Chen, W.L.**, Alvarez, R., Park, S. “Layer-Wise Distillation for Efficient Ranking Models.” NeurIPS 2023.
- Chen, W.L.**, Nakamura, T. “Revisiting Quantization Error in Sparse Transformers.” ICML 2023.
- Park, S., **Chen, W.L.** “Calibrated Offline Evaluation for Search Ranking.” EMNLP 2022.
- Chen, W.L.**, Duarte, M., Alvarez, R. “Representation Collapse Under Aggressive Pruning.” NeurIPS 2021.

CERTIFICATIONS

-
- NVIDIA Deep Learning Institute – Fundamentals of Accelerated Computing with CUDA (2022)